Chapter 5, Section 1

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1. Let C, D be any two divisors on a surface X, and let the corresponding invertible sheaves be $\mathcal{L}$ and $\mathcal{M}$. Show that
$$C.D = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}).$$

Proof. Applying the Riemann-Roch theorem to the divisor $-C - D$ gives

$$\chi(\mathcal{O}_X(-C-D)) = \frac{1}{2}(-C-D).(-C-D-K) + \chi(\mathcal{O}_X)$$

The first term on the right hand side simplies to

$$(-C-D).(-C-D-K) = C^2 + 2C.D. + D^2 + C.K + D.K.$$

Substitutting back to above and isolating $C.D$ gives

$$\begin{aligned} C.D &= -\chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \frac{1}{2}C.(C+K) - \frac{1}{2}C.(C+K) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}C.(-C-K)\right) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}D.(-D-K)\right) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \chi(\mathcal{O}_X(-C)) - \chi(\mathcal{O}_X(-D)). \end{aligned}$$

□

2. Let H be a very ample divisor on the surface X, corresponding to a projective embedding $X \subseteq \mathbb{P}^N$. If we write the Hilbert polynomial of X (III, Ex. 5.2) as

$$P(z) = \frac{1}{2}az^2 + bz + c,$$

show that $a = H^2$, $b = \frac{1}{2}H^2 + 1 - \pi$, where π is the genus of a nonsingular curve representing H, and $c = 1 + p_a$. Thus the *degree* of X in $\mathbb{P}^N$, as defined in (I, §7), is just H^2. Show also that if C is any curve in X, then the degree of C in $\mathbb{P}^N$ is just $C.H$.

Proof. By the Riemann-Roch theorem, we have the equality

$$P(z) = \frac{1}{2}H^2z^2 - \frac{1}{2}(H.K)z + 1 + p_a.$$

Matching coefficients readily gives a, b, and c. We can rewrite the coefficient of z as

$$-\frac{1}{2}(H.K) = \frac{1}{2}H^2 - \frac{1}{2}H.(H+K),$$

and the adjunction formula (1.5) gives

$$2\pi - 2 = H.(H+K).$$

We conclude that
$$b = \frac{1}{2}H^2 + 1 - \pi.$$

If C is any curve in X, then the degree of C in $\mathbb{P}^N$ is just the degree of $\mathcal{O}_X(H) \otimes \mathcal{O}_C$ as a line bundle on C, which is equal to $C.H$ by (1.3). □

3. Recall that the *arithmetic genus* of a projective scheme D of dimension 1 is defined as $p_a(D) = 1 - \chi(\mathcal{O}_D)$ (III, Ex. 5.3).

(a) If D is an effective divisor on a surface X, use (1.6) to show that $2p_a(D) - 2 = D.(D+K)$.

(b) $p_a(D)$ depends only on the linear equivalence class of D on X.

(c) More generally, for *any* divisor D on X, we define the *virtual arithmetic genus* (which is equal to the ordinary arithmetic genus if D is effective) by the same formula: $2p_a - 2 = D.(D+K)$. Show that for any two divisors C, D, we have

$$p_a(-D) = D^2 - p_a(D) + 2,$$

and

$$p_a(C+D) = p_a(C) + p_a(D) + C.D - 1.$$

Proof.

(a) We have the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

and taking Euler characteristics gives

$$\chi(\mathcal{O}_D) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)).$$

On the other hand, the Riemann-Roch theorem gives

$$\chi(\mathcal{O}_X(-D)) = \frac{1}{2}D.(D+K) + \chi(\mathcal{O}_X).$$

Substituting gives

$$1 - \chi(\mathcal{O}_D) = 1 + \frac{1}{2}D.(D+K).$$

(b) Follows from (a) and (1.1(4)), which states $D.(D+K)$ depends only on the linear equivalence class of D on X.

(c)

$$\begin{aligned}
2p_a(-D) - 2 &= -D.(-D+K) \\
&= D.(D-K) \\
&= 2D^2 - D.(D+K) \\
&= 2D^2 - 2p_a(D) + 2 \\
2p_a(C+D) - 2 &= (C+D).(C+D+K) \\
&= C.(C+K) + D.(D+K) + 2(C.D) \\
&= 2p_a(C) - 2 + 2p_a(D) - 2 + 2(C.D)
\end{aligned}$$

□

4. (a) If a surface X of degree d in $\mathbb{P}^3$ contains a straight line $C = \mathbb{P}^1$, show that $C^2 = 2 - d$.

(b) Assume $\operatorname{char} k = 0$, and show for every $d \geqslant 1$, there exists a nonsingular surface X of degree d in $\mathbb{P}^3$ containing the line $x = y = 0$.

Proof.

(a) By (Ex. 1.5), $C^2 = -C.K - 2$. By (1.3), $C.K = d - 4$, because C is a straight line and $\mathcal{O}_X(K) = \mathcal{O}_X(d-4)$. Hence, $C^2 = 2 - d$.

(b) We can prove the following more general statement:

Proposition 0.1. *For every $n \geqslant 3$ and $d > 0$, and for every linear subspace $H \subset \mathbb{P}^n$ of codimension at least 2, there exists a nonsingular hypersurface of degree d in $\mathbb{P}^n$ containing H.*

We can assume H has codimension two and is defined by the vanishing of $x = y = 0$. Bertini's theorem (III, 10.9.2) says a general hypersurface X of degree d in $\mathbb{P}^3$ containing H is nonsingular away from H. Also, any such X is is defined by the vanishing of $xF + yG = 0$, where F, G define surfaces of degree $d-1$. If P is a point in H, then X is singular at P if and only if both F, G vanish at P. This is a closed condition, so we can find plenty of choices of F, G which give a nonsingular hypersurface containing H.

Example 0.2. For $n = 3$ and $d \geqslant 2$, the surface defined by

$$x(z^{d-1} + x^{d-1}) + y(w^{d-1} + y^{d-1}) = 0$$

is nonsingular and contains L.

□

5. (a) If X is a surface of degree d in $\mathbb{P}^3$, then $K^2 = d(d-4)^2$.

(b) If X is a product of two nonsingular curves C, C', of genus g, g' respectively, then $K^2 = 8(g-1)(g'-1)$.

Proof.

(a) The adjunction formula gives $\mathcal{O}_X(K) = \mathcal{O}_X(d-4)$. If H is a general hyperplane section of X in $\mathbb{P}^3$, then $K = (d-4)H$, and $H^2 = d$ by (Ex. 1.2). Hence,

$$K^2 = (d-4)^2 H^2 = d(d-4)^2.$$

(b) By (II, Ex. 8.3), there is an isomorphism $\omega_X \cong p_1^*\omega_C \otimes p_2^*\omega_{C'}$. In particular, if $\mathfrak{k}'$ and $\mathfrak{k}'$ are canonical divisors of C and C', respectively, then $p_1^*\mathfrak{k} + p_2^*\mathfrak{k}'$ is a canonical divisor of X. Denote $D = p_1^*\mathfrak{k}$ and $D' = p_2^*\mathfrak{k}'$. The support of D consists of $2g-2$ copies of C' which are the fibres of the points in $\mathfrak{k}$ with respect to $p_1 : X \to C$. It follows that

$$D.D' = (\deg K_C) \cdot (\deg K_{C'}) = 4(g-1)(g'-1).$$

Also, we can always find two canonical divisors with disjoint support, and since the intersection pairing only depends on the linear equivalence class, we have $D^2 = D'^2 = 0$. We conclude that $K^2 = 2D.D' = 8(g-1)(g'-1)$.

□

6. (a) If C is a curve of genus g, show that the diagonal $\Delta \subseteq C \times C$ has self-intersection $\Delta^2 = 2-2g$. (Use the definition of ω_C in (II, §8).)

(b) Let $l = C \times \mathrm{pt}$ and $m = \mathrm{pt} \times C$. If $g \geqslant 1$, show that l, m, and Δ are linearly independent in $\operatorname{Num}(C \times C)$. Thus $\operatorname{Num}(C \times C)$ has rank $\geqslant 3$, and in particular, $\operatorname{Pic}(C \times C) \neq p_1^* \operatorname{Pic} C \oplus p_2^* \operatorname{Pic} C$.

Proof.

(a) By definition, $\omega_C \cong \mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta$, and by (1.3), $\deg_\Delta(\mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta) = -\Delta^2$.

(b) Suppose $D = al + bm + c\Delta = 0$ in $\operatorname{Num}(C \times C)$ for some integers $a, b, c \in \mathbb{Z}$. It is not hard to see

$$l^2 = m^2 = 0, \quad l.m = 1, \quad \Delta.l = \Delta.m = 1,$$

which implies

$$D.m = b + c = 0, \quad D.l = a + c = 0, \quad D.\Delta = a + b + c(2-2g) = 0.$$

Combining the first two gives $a + b + 2c = 0$. The condition $g \geqslant 1$ forces $a, b, c = 0$, which is what we wanted to show.

□

7. *Algebraic Equivalence of Divisors.* Let X be a surface. Recall that we have defined an algebraic family of effective divisors on X, parametrized by a nonsingular curve T, to be an effective Cartier divisor D on $X \times T$, flat over T (III, 9.8.5). In this case, for any two closed points $0, 1 \in T$, we say that the corresponding divisors D_0, D_1 on X are prealgebraically equivalent. Two arbitrary divisors are prealgebraically equivalent if they are differences of prealgebraically equivalent divisors. Two divisors D, D' are *algebraically equivalent* if there is a finite sequence $D = D_0, D_1, \dots, D_n = D'$ with D_i and D_{i+1} prealgebraically equivalent for each i.

(a) Show that the divisors algebraically equivalent to 0 form a subgroup of $\operatorname{Div} X$.

(b) Show that linearly equivalent divisors are algebraically equivalent.

(c) Show that algebraically equivalent divisors are numerically equivalent.

Proof.

(a) A divisor D is prealgebraically equivalent to 0 if and only if $D = E - E'$, where E and E' are prealgebraically equivalent effective divisors. Thus, D is algebraically equivalent to 0 if and only if we can write $D = \sum(E_i - E_i')$, where E_i and E_i' are prealgebraically equivalent effective divisors for each i. If $D' = \sum(F_j - F_j')$ is another divisor algebraically equivalent to 0, then

$$D - D' = \sum(E_i - E_i') + \sum(F_j' - F_j),$$

so $D - D'$ is also algebraically equivalent to 0. Hence, the divisors algebraically equivalent to 0 form a subgroup of $\operatorname{Div} X$.

(b) We want to show any principal divisor $D = (f)$ is algebraically equivalent to 0, where f is some rational function on X. Then we can write $D = E - E'$, where E and E' are effective divisors, and f can locally be expressed as a quotient g/h, where g and h are locally defining equations for E and E' as Cartier divisors. We will show E and E' are prealgebraically equivalent. Consider the effective Cartier divisor $\tilde{D}$ on $X \times \mathbb{A}^1$ defined by $F = (1-t)g + th$. If $f = g'/h'$ is any another representative so that $g'h = gh'$, define $F' = (1-t)g' + th'$. To show H is well-defined, it is enough to show $F/F' \in \mathcal{O}^*_{X \times \mathbb{A}^1}$. It is easily check that

$$\frac{(1-t)g + th}{(1-t)g' + th'} \cdot \frac{h}{h} = \frac{(1-t)gh + th^2}{(1-t)gh' + thh'} = \frac{h}{h'} \in \mathcal{O}^*_X,$$

hence $F/F' \in \mathcal{O}^*_{X \times \mathbb{A}^1}$. Notice that $\tilde{D}$ is flat over $\mathbb{A}^1$ since it is surjective onto $\mathbb{A}^1$ (III, 9.7), and the divisors on X corresponding to the fibres over $t = 0, 1$ are E, E', respectively. Hence, E and E' are algebraically equivalent.

(c) We want to show if any divisor D on X algeberaically equivalent to 0 is numerically equivalent to 0. Since divisors algebraically equivalent to zero can be written as a sum of differences of prealgebraically equivalent to 0, it is enough to assume $D = E_0 - E_1$, where E_0 and E_1 are prealgebraically equivalent effective divisors. Let T be a nonsingular irreducible curve, and let E be an effective Cartier divisor on $X \times T$, flat over T, such that the divisors corresponding to the fibres t are $E_t \subset X$. It is enough to show for any irreducible curve C in X, such that C is transveral to E_0 and E_1, we have $C.E_0 = C.E_1$. Since X is a projective scheme, we can view $X \times T$ as a closed subscheme of $\mathbb{P}^n_T$ for some $n > 0$, as well as $C \times T \subset X \times T$. The hypothesis of (9.9) is satisfied for $C \times T$ since $C \times T \to T$ is a flat morphism, so applying (9.9) to the invertible sheaf $\mathcal{L} = \mathcal{O}_{X \times T}(E) \otimes \mathcal{O}_{C \times T}$ on $C \times T$ shows the Hilbert polynomial $P_t(z)$ of $\mathcal{L}_t = \mathcal{O}_{X \times T}(E) \otimes \mathcal{O}_{C \times t}$ is independent of t. Notice that $C.E_t = \deg \mathcal{L}_t$ for $t = 0, 1$ by (1.3), where $\deg \mathcal{L}_t$ is the leading coefficient of $P_t(z)$. Hence, $C.E_t$ is independent of t for $t = 0, 1$, so D is numerically equivalent to zero.

□

8. *Cohomology Class of a Divisor.* For any divisor D on the surface X, we define its cohomology class $c(D) \in H^1(X, \Omega_X)$ by using the isomorphism $\operatorname{Pic} X \cong H^1(X, \mathcal{O}_X^*)$ of (III, Ex. 4.5) and the sheaf homomorphism $d\log : \mathcal{O}_X^* \to \Omega_X$ (III, Ex. 7.4c). Thus we obtain a group homomorphism $c : \operatorname{Pic} X \to H^1(X, \Omega_X)$. On the other hand, $H^1(X, \Omega_X)$ is dual to itself by Serre duality (III, 7.13), so we have a nondegenerate bilinear map

$$\langle \quad , \quad \rangle : H^1(X, \Omega_X) \times H^1(X, \Omega_X) \to k.$$

(a) Prove that this is compatible with the intersection pairing, in the following sense: for any two divisors D, E on X, we have

$$\langle c(D), c(E) \rangle = (D.E) \cdot 1$$

in k.

(b) If $\operatorname{char} k = 0$, use the fact that $H^1(X, \Omega_X)$ is a finite-dimensional vector space to show that $\operatorname{Num} X$ is a finitely generated free abelian group.

Proof.

(a) We first consider the case when D and E are nonsingular irreducible curves meeting transversally. The pairing of cohomology is defined as the following: the natural map $\Omega_X \otimes \mathcal{O}_D \to \Omega_D$ induces a linear map in cohomology $H^1(X, \Omega_X) \to H^1(D, \Omega_D)$. Since Ω_D is a dualizing sheaf for D, we have a trace map $t_D : H^1(D, \Omega_D) \to k$. Composing, we obtain a linear map $\theta_D : H^1(X, \Omega_X) \to k$. If E is another irreducible divisor of X, then we can consider its image in $H^1(X, \Omega_X)$ under c. Hence, $\langle c(D), c(E) \rangle = \theta_D(c(E)) = \theta_E(c(D))$.

Generalizing, we want to show the following diagram commutes

$$\begin{tikzcd} \operatorname{Pic}(X) \arrow[r, "(D.\cdot)\cdot 1"] \arrow[d, "c"'] & \mathbb{Z} \arrow[d] \\ H^1(X, \Omega_X) \arrow[r, "\theta_D"] & k \end{tikzcd}.$$

The top row factos into

$$\begin{tikzcd} & \operatorname{Pic} D \arrow[d, "\deg"] \\ \operatorname{Pic} X \arrow[ur] \arrow[r, "(D.\cdot)\cdot 1"] & \mathbb{Z} \end{tikzcd}.$$

The bottom row factors into

$$\begin{tikzcd} & H^1(D, \Omega_D) \arrow[d, "t_D"] \\ H^1(X, \Omega_X) \arrow[ur] \arrow[r, "\theta_D"] & k \end{tikzcd}.$$

The left column factors into

$$\begin{tikzcd} \operatorname{Pic} X \arrow[r] \arrow[d, "c"] & \operatorname{Pic} D \arrow[d, "c"] \\ H^1(X, \Omega_X) \arrow[r] & H^1(D, \Omega_D) \end{tikzcd}.$$

The result of (III, Ex. 7.4) gives the commutative diagram

$$\begin{tikzcd} \operatorname{Pic} D \arrow[r, "\deg"] \arrow[d, "c"'] & \mathbb{Z} \arrow[d] \\ H^1(D, \Omega_D) \arrow[r, "t_D"] & k \end{tikzcd}.$$

So the original diagram commutes. For arbitrary divisor D, we can always write D as a sum of irreducible curves.

(b) We have an injection $c : \operatorname{Num} X \hookrightarrow H^1(X, \Omega_X)$, and in characteristic 0, this means $k \otimes_{\mathbb{Z}} \operatorname{Num} X \hookrightarrow H^1(X, \Omega_X)$ is also an injection. Since $H^1(X, \Omega_X)$ is a finite-dimensional vector space, this means $\operatorname{Num} X$ is finitely generated. Also, $\operatorname{Num} X$ has no torsion, so $\operatorname{Num} X$ is actually a free finitely generated $\mathbb{Z}$-module.

$\square$

9. (a) If H is an ample divisor on the surface X, and if D is any divisor, show that

$$(D^2)(H^2) \leqslant (D.H)^2.$$

(b) Now let X be a product of two curves $X = C \times C'$. Let $l = C \times \text{pt}$, and $m = \text{pt} \times C'$. For any divisor D on X, let $a = D.l$, $b = D.m$. Then we say D has *type* (a, b). If D has type (a, b), with $a, b \in \mathbb{Z}$, show that

$$D^2 \leqslant 2ab,$$

and equality holds if and only if $D = bl + am$.

Proof.

(a) By (1.9), (1.9.1), and (Ex. 1.8), the satement reduces to the Cauchy-Schwartz inequality for Lorenztian inner products.

(b) By Nakai-Moishezon (1.10), $H = l + m$ is ample, with $H^2 = 2$. Apply the previous exercise to $D' = D + al + bm$ and H, which simplifies to

$$D'^2 \leqslant 2(a+b)^2.$$

Expanding the left-hand side gives

$$D'^2 = D^2 + 2a^2 + 2b^2 + 2ab.$$

Substitutting and simplifying gives

$$D^2 \leqslant 2ab.$$

□

Example 0.3. Consider the quadric surface $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and the divisor H of type $(1, 1)$. If D be a divisor of type (a, b), then by (1.4.3), the left-hand side is equal to $2ab$, while the right-hand side is equal to $(a + b)^2$ (IV, 6.4.1) (Ex. 1).

10. *Weil's Proof [2] of the Analogue of the Riemann Hypothesis for Curves.* Let C be a curve of genus g defined over the finite field $\mathbb{F}_q$, and let N be the number of points of C rational over $\mathbb{F}_q$. Then $N = 1 - a + q$, with $|a| \leqslant 2g\sqrt{q}$. To prove this, we consider C as a curve over the algebraic closure k of $\mathbb{F}_q$. Let $f : C \to C$ be the k-linear Frobenius morphism obtained by taking qth powers, which makes sense since C is defined over $\mathbb{F}_q$, so $X_q \cong X$ (IV, 2.4.1). Let $\Gamma \subseteq C \times C$ be the graph of f, and let $\Delta \subseteq C \times C$ be the diagonal. Show that $\Gamma^2 = q(2 - 2g)$, and $\Gamma.\Delta = N$. Then apply (Ex. 1.9) to $D = r\Gamma + s\Delta$ for all r and s to obtain the result.

Proof. Applying the adjunction formula (1.5) to Γ gives

$$\Gamma^2 = 2g - 2 - \Gamma.K,$$

where K is a canonical divisor of $C \times C$. We can choose K to be $K = p_1^*K_C + p_2^*K_C$, where K_C is a canonical divisor of C. Each $p_i^*K_C$ is a sum of $2g - 2$ lines, either $l = C \times \text{pt}$ or $m = \text{pt} \times C$ depending on $i = 1, 2$. Thus, $K = (2g - 2)(l + m)$. Since Γ is the graph of f, $\Gamma.l = \deg f = q$ and $\Gamma.m = 1$.

Now, we can compute $\Gamma.K$

$$\Gamma.K = (2g - 2)(\Gamma.(l + m)) = (2g - 2)(q + 1).$$

Plugging into (1.5) gives

$$\Gamma^2 = 2g - 2 - \Gamma.K = q(2 - 2g).$$

Notice that $\Gamma \cap \Delta$ set-theoretically consists of all the closed points of (P, P) such that $P \in C$ is a fixed point of f. By (1.4), we conclude that $\Gamma.\Delta = N$.

We have

$$D.l = r\Gamma.l + s\Delta.l = rq + s, \quad D.m = r\Gamma.m + s\Delta.m = r + s.$$

By (Ex. 1.8), we have

$$D^2 \leqslant 2(rq+s)(r+s).$$

Expanding the left-hand side gives

$$D^2 = r^2\Gamma^2 + 2rs\Gamma.\Delta + s^2\Delta^2 = r^2q(2-2g) + 2rsN + s^2(2-2g).$$

Hence, we have

$$r^2q(2-2g) + 2rsN + s^2(2-2g) \leqslant 2(rq+s)(r+s).$$

Plugging in $N = 1-a+q$ and simplifying in a similar manner to (IV, Ex. 4.16) gives the inequality

$$|a| \leqslant |\theta q g + g/\theta|$$

for all $\theta \in \mathbb{R}$. The local minimum and maximum of the right-hand side is $\pm 2g\sqrt{q}$. Hence, $|a| \leqslant 2g\sqrt{q}$. □